

AI-Powered Enterprise E-Commerce Platform

Functional Requirements Specification (FRS)

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Table of Contents

- 1. Introduction
 - 2. Purpose
 - 3. System Overview
 - 4. Functional Requirement Categories
 - 5. User Roles
 - 6. Detailed Functional Requirements
 - 7. System Inputs and Outputs
 - 8. Functional Requirement Priority
 - 9. Assumptions
 - 10. Dependencies
 - 11. Traceability Matrix
 - 12. References
-

1. Introduction

This document describes the functional requirements of the AI-Powered Enterprise E-Commerce Platform.

Functional requirements explain how the system should behave and what services it should provide to users.

These requirements will guide:

- system design,
- database design,

- API development,
 - AI integration,
 - testing activities,
 - and future maintenance.
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2. Purpose

The purpose of this document is to:

- define system functionalities,
 - identify user interactions,
 - establish system behavior,
 - reduce misunderstandings,
 - support development activities.
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3. System Overview

The proposed system is an intelligent e-commerce platform that combines traditional e-commerce services with Artificial Intelligence technologies.

The platform consists of:

Customer Module

Administration Module

Inventory Module

Order Module

AI Module

Analytics Module

Security Module

4. Functional Requirement Categories

The requirements are grouped into the following categories:

1. User Management
2. Authentication
3. Product Management
4. Shopping Cart
5. Order Management

6. Payment Management
 7. Inventory Management
 8. Recommendation System
 9. Fraud Detection
 10. Analytics
 11. AI Assistant
 12. Notifications
 13. Reporting
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5. User Roles

The following user roles have been identified.

Customer

Uses the platform to browse and purchase products.

Administrator

Manages the entire platform.

Inventory Manager

Manages products and stock levels.

Business Owner

Uses reports and analytics.

Delivery Personnel

Handles product delivery activities.

6. Detailed Functional Requirements

6.1 User Registration Module

FR-001 User Registration

The system shall allow users to create accounts.

Input:

- Name
- Email
- Password
- Phone Number

Output:

- Successful account creation message.

Priority:

High

FR-002 Email Validation

The system shall validate email formats.

Priority:

High

FR-003 Duplicate Account Prevention

The system shall prevent duplicate registrations.

Priority:

High

FR-004 Password Encryption

The system shall securely store passwords.

Priority:

Critical

6.2 Authentication Module

FR-005 User Login

The system shall allow registered users to login.

Priority:

Critical

FR-006 Password Reset

The system shall allow password recovery.

Priority:

Medium

FR-007 Logout

The system shall allow users to logout securely.

Priority:

High

FR-008 Role-Based Access Control

The system shall provide different permissions for:

- Customers
- Administrators
- Inventory Managers
- Business Owners

Priority:

Critical

6.3 User Profile Module

FR-009 Profile Management

Users shall be able to:

- update profile information,
- change passwords,
- manage addresses.

Priority:

High

6.4 Product Management Module

FR-010 Product Creation

Administrators shall be able to add products.

Priority:

Critical

FR-011 Product Update

Administrators shall update product information.

Priority:

Critical

FR-012 Product Deletion

Administrators shall remove products.

Priority:

High

FR-013 Product Categories

The system shall support product categorization.

Priority:

High

FR-014 Product Images

The system shall support multiple product images.

Priority:

Medium

FR-015 Product Search

Customers shall search products.

Priority:

Critical

FR-016 Product Filtering

Customers shall filter products using:

- price,
- category,
- ratings,
- popularity.

Priority:

High

FR-017 Product Reviews

Customers shall submit reviews and ratings.

Priority:

Medium

6.5 Shopping Cart Module

FR-018 Add to Cart

Customers shall add products to cart.

Priority:

Critical

FR-019 Remove from Cart

Customers shall remove products.

Priority:

High

FR-020 Quantity Management

Customers shall modify quantities.

Priority:

High

FR-021 Cart Summary

The system shall display:

- subtotal,
- taxes,
- discounts,
- total amount.

Priority:

High

6.6 Wishlist Module

FR-022 Wishlist Management

Customers shall save products for future purchases.

Priority:

Medium

6.7 Order Management Module

FR-023 Checkout Process

Customers shall place orders.

Priority:

Critical

FR-024 Order Confirmation

The system shall generate confirmations.

Priority:

Critical

FR-025 Order Tracking

Customers shall track order status.

Priority:

High

FR-026 Order Cancellation

Customers shall cancel orders.

Priority:

Medium

FR-027 Order History

Customers shall view previous orders.

Priority:

High

6.8 Payment Module

FR-028 Payment Processing

The system shall process payments.

Priority:

Critical

FR-029 Refund Processing

The system shall support refunds.

Priority:

Medium

FR-030 Payment History

Users shall view payment history.

Priority:

Medium

6.9 Inventory Module

FR-031 Stock Management

Inventory managers shall manage stock.

Priority:

Critical

FR-032 Automatic Stock Update

Inventory shall update after purchases.

Priority:

Critical

FR-033 Low Stock Alerts

The system shall notify administrators.

Priority:

High

6.10 Recommendation System

FR-034 Personalized Recommendations

The system shall recommend products using:

- purchase history,
- browsing behavior,
- customer preferences.

Priority:

High

FR-035 Similar Product Suggestions

The system shall display similar products.

Priority:

Medium

FR-036 Trending Products

The system shall identify trending items.

Priority:

Medium

6.11 Fraud Detection Module

FR-037 Fraud Monitoring

The system shall monitor suspicious activities.

Priority:

Critical

FR-038 Fraud Alerts

Administrators shall receive alerts.

Priority:

Critical

FR-039 Transaction Risk Score

Transactions shall receive risk scores.

Priority:

High

6.12 Customer Segmentation Module

FR-040 Customer Clustering

The system shall group customers based on behavior.

Priority:

Medium

FR-041 Customer Analytics

Administrators shall view customer segments.

Priority:

Medium

6.13 Sales Forecasting Module

FR-042 Demand Forecasting

The system shall predict future product demand.

Priority:

High

FR-043 Sales Forecast Reports

Administrators shall view forecasting reports.

Priority:

Medium

6.14 AI Assistant Module

FR-044 Customer Chat Assistant

The system shall answer customer questions.

Priority:

Medium

FR-045 Administrative Assistant

The system shall answer business questions.

Example:

- Which products generated the highest revenue?

- What are expected sales next month?

Priority:

Medium

6.15 Notification Module

FR-046 Email Notifications

The system shall send notifications.

Examples:

- Order confirmation
- Password reset
- Shipping updates

Priority:

High

FR-047 In-System Notifications

Users shall receive notifications inside the platform.

Priority:

Medium

6.16 Reporting Module

FR-048 Sales Reports

The system shall generate sales reports.

Priority:

High

FR-049 Customer Reports

The system shall generate customer analytics reports.

Priority:

Medium

FR-050 Inventory Reports

The system shall generate inventory reports.

Priority:

Medium

7. System Inputs

Examples:

- User information
 - Product information
 - Order information
 - Payment information
 - Customer behavior data
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8. System Outputs

Examples:

- Recommendations
 - Reports
 - Notifications
 - Analytics Dashboards
 - Forecast Results
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9. Functional Requirement Priority Levels

Level	Description
Critical	System cannot function without it
High	Very important functionality
Medium	Useful but not essential
Low	Future enhancement

10. Assumptions

- 1. Internet connectivity is available.
- 2. Users possess basic computer knowledge.
- 3. Product data is maintained properly.
- 4. Sufficient historical data exists for AI models.

11. Dependencies

The system depends on:

- Database services
- Email services
- AI services
- Payment providers
- Notification services

12. Requirement Traceability Matrix

Business Requirement	Functional Requirement
Increase sales	FR-034, FR-042
Improve customer satisfaction	FR-015, FR-034
Reduce fraud	FR-037, FR-038
Improve inventory management	FR-031, FR-042
Improve decision making	FR-043, FR-048

13. Conclusion

This document defines all major functionalities required by the AI-Powered Enterprise E-Commerce Platform.

These requirements provide the foundation for:

- system architecture,
- database design,
- API development,
- AI integration,
- testing,
- deployment,
- and future maintenance.

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